

Sustainability (Environmental Impact of AI)

This session addressed the **environmental impact of AI** and the balance between innovation, energy consumption, and sustainability.

Key Findings from Faculty Discussion

- **High Energy Costs:** Faculty recognized the significant energy demands of AI hardware and software..
- **Corporate Commitments:** Participants acknowledged that major companies like Microsoft and Google have made strides toward carbon neutrality, while others; such as Grok appear less committed or transparent about environmental impact.
- **Educational Relevance:** Despite concerns about energy use, some faculty agreed they would not factor environmental cost into their decision to adopt AI in teaching, describing education as a negligible sector in overall AI energy consumption. Faculty emphasized that the benefits of AI integration outweigh environmental costs,

References

Artificial Intelligence for Sustainability: A Systematic Review (MDPI, 2025) ([link](#))

A comprehensive review analyzing how AI supports climate action, resource optimization, and sustainable development while acknowledging its ecological trade-offs.

Artificial Intelligence in Sustainable Development Research (Nature, 2025) ([link](#))

Synthesizing over 700 studies, this article shows how AI accelerates sustainability research by enabling predictive modeling for environmental systems and renewable energy.

Using AI to Improve Energy Efficiency for Campuses Nationwide (University of Missouri, 2025) ([link](#))

Mizzou researchers developed AI-driven systems that optimize campus energy management, cutting operational costs and reducing carbon emissions.

How Hungry is AI? Benchmarking Energy, Water, and Environmental Footprint (arXiv, 2025) ([link](#))

This benchmark reveals significant variation in AI model energy consumption—some large models consume over 70× more power than lightweight architectures.

The Energy and Environmental Footprint of AI (Vanderbilt Law Research Paper, 2025) ([link](#))

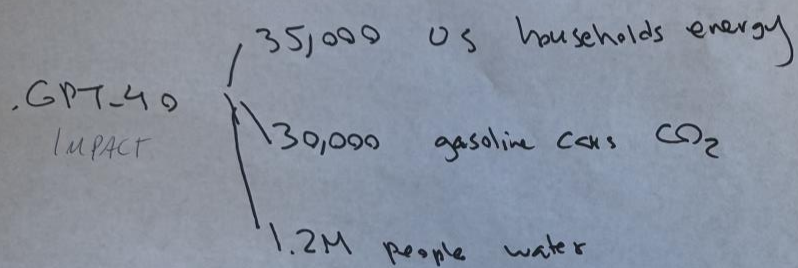
Analyzes discrepancies between AI carbon calculators, finding that estimates for identical tasks can differ by more than 50×, complicating sustainability reporting.

Life-Cycle Emissions of AI Hardware: A Cradle-to-Grave Approach (arXiv, 2025) ([link](#))

Provides a full life-cycle assessment of AI accelerators, showing that despite generational efficiency gains, hardware production remains a major emissions source.

SUSTAINABILITY

AI enables predictive modeling for renewable energy, water consumption, etc



Should we factor AI's energy use into our teaching?

No, education is a negligible sector in AI energy use.

→ the benefits outweigh the costs!